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A physics based retrieval and quality assessment of bathymetry from suboptimal hyperspectral data

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ABSTRACT

In order to retrieve bathymetry, substratum type and the concentrations of the optically active constituents of the water column, an integrated physics based mapping approach was applied to airborne hyperspectral data of Moreton Bay, Australia. The remotely sensed data were sub-optimal due to high and mid-level cloud covers. Critical to the correct interpretation of the resultant coastal bathymetry map was the development of a quality control procedure based on additional outputs of the integrated physics based mapping approach and the characteristics of the instrument. These two outputs were: an optical closure term which defines differences between the image and model based remote sensing signal; and an estimate of the relative contribution of the substratum signal to the remote sensing signal. This quality control procedure was able to identify those pixels with a reliable retrieval of depth and to detect thin and thick clouds and their shadows, which were subsequently masked out from further analysis. The derived coastal bathymetry in depths ranging 4–13 m for the mapped area was within ±15% of boat-based multi-beam acoustic mapping survey of the same area. The agreement between the imaging spectrometry and the acoustic datasets varies as a function of the contribution of the bottom visibility to the remote sensing signal. As expected, there was greater agreement in shallower clear water (±0.67 m) than quasi-optically deep water (±1.35 m). The quantitative identification and screening of the optically deep waters and the quasi-optically deep waters led to improved precision in the depth retrieval. These results suggest that the physics based mapping approach adopted in this study performs well for retrieving water column depths in coastal waters in water depths ranging 4–13 m for the area and conditions studied, even with sub-optimal imagery.

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1. Introduction

The ability to accurately estimate water column depth over large areas and/or in remote locations is directly relevant to environmental management, exploration, defence and research applications. The potential to utilise remote sensing to this end has been discussed for over 2 decades, but has been mostly limited to empirical approaches (Clark et al., 1987; Lyzenga, 1981; Philpot, 1989) that are not easily transferable across study areas or data types (Lee et al., 2001; Stumpf et al., 2003). Recently, approaches for mapping bathymetry in optically shallow water bodies have evolved to non-linear optimization of semi-analytical models (Adler-Golden et al., 2005; Albert & Gege, 2006; Lee et al., 1999, 2001) and comparative methods of spectral library matching (Louchard et al., 2003; Mobley et al., 2005) from hyperspectral data and modelled data. In some cases these approaches can also be used to produce corrected substratum reflectance spectra and to quantify concentration of organic and inorganic water constituents.

Lee et al. (1999, 1998) developed a semi-analytical model for shallow water remote sensing based on the analytical model proposed by Maritorena et al. (1994). Lee et al. (1999, 2001) used an inversion-optimization approach to simultaneously derive water depth and water column properties from hyperspectral data in coastal waters. Adler-Golden et al. (2005) present an algorithm similar to that of Lee et al. (2001). However, it makes the simplifying assumption of constant water optical properties within the scene. McIntyre et al. (2006) presented an application of the Lee et al. (2001) inversion modelling approach to clear waters which included a quantitative comparison of model-derived depth with high resolution multi-beam acoustic bathymetry data.

Several authors recently extended the method developed by Lee et al. (1999, 2001) by incorporating linear un-mixing of the benthic cover. Giardino et al. (2007) used two substrate classes (bare sand and submerged macrophytes) for the littoral zone of a lake, while Goodman and Ustin (2007) and Klonowski et al. (2007) integrated a semi-

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analytical inversion model with a linear un-mixing of three substratum types for coral reef environments.

In this work, an enhanced implementation of the inversion/optimization approach by Lee et al. (1999, 2001) was applied to airborne hyperspectral data of a shallow water coastal embayment. The approach was used to estimate bathymetry, substratum composition (i.e. fractional cover of e.g. sand, silt, mud, seagrass and macroalgae) and the concentrations of the optically active constituents of the water column, including chlorophyll, coloured dissolved organic matter (CDOM) and non algal particulate matter (NAP). Fig. 1 presents the schematic flowchart of the integrated physics based mapping approach that includes atmospheric correction and an objective process of quality control.

To overcome the inadequate quality of portions of the imagery due the cloud cover in the area, a quality control procedure based on physical quantities was developed in order to identify those pixels where reliable retrieval of depth could be performed. The quality control procedure was based on a measure of optical closure, i.e., the similarity between below-surface modelled reflectance and that measured in the image, and a quantitative estimate of the contribution of the substratum signal to the remote sensing signal.

The comparison of the depth retrieved from the hyperspectral imagery with the depth measured during a boat-based multi-beam acoustic mapping survey demonstrates the effect of this objective quality control procedure. The accuracy and precision of the coastal bathymetry retrieval is discussed as a function of the quantitative estimate of the contribution of the substratum to the remote sensing signal.

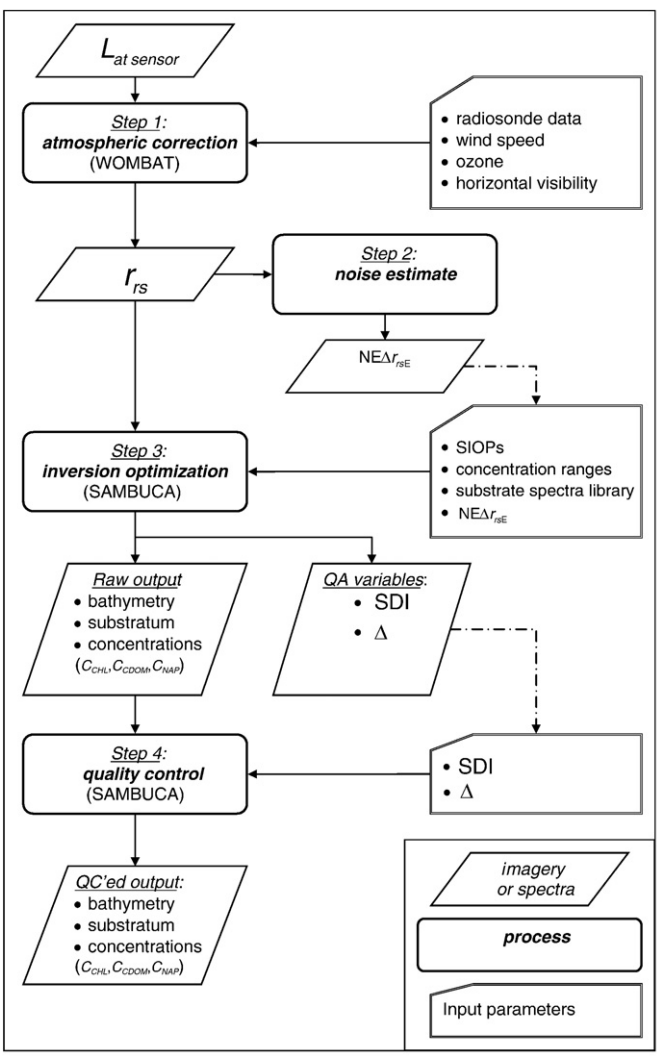

Fig. 1. Concept and schematic flow chart of the integrated physics based mapping approach for estimating bathymetry, substratum composition (i.e. fractional cover of e.g. sand, silt, mud, seagrass, macroalgae) and the concentrations of the optically active constituents of the water column (C_{CHL}, C_{CDOM}, C_{NAP}). All symbols are defined in Table 1.

2. Data and methodology

2.1. Study site

The study site is located in Moreton Bay (27°30′S, 153°30′E), a large embayment located on the east coast of Australia. The Bay is surrounded by shallow banks to the north and protected by Moreton and North Stradbroke Islands on the east and southeast sides (Fig. 2). Moreton Bay can be considered a representative example of the range of water quality and substratum cover types typically found in coastal and coral reef environments. (Phinn et al., 2005, 2008).

Moreton Bay substratum contains significant areas of unconsolidated sediments, ranging from fine-silt muds in the western bay to silicate sands in the eastern bay. Extensive seagrass beds and macroalgae occur throughout the bay, as do bedrock outcrops and fringing reefs. Due to the number of creeks and rivers that drain into the western part of the Bay and the oceanic openings on its eastern side, the water column usually ranges from freshwater influenced, and often turbid in the western bay, to oceanic water dominated and clear blue-green waters of the eastern bay (Phinn et al., 2005, 2008).

2.2. Image acquisition and processing

2.2.1. CASI-2 airborne hyperspectral Images

Airborne hyperspectral imagery was acquired on 28 July 2004 over the Eastern Banks area in Moreton Bay (Fig. 2A) with a CASI-2 (Compact Airborne Spectrographic Imager) measuring upwelling radiance [$\mu Wcm^{-2}sr^{-1}nm^{-1}$]. The hyperspectral dataset was collected as part of a larger project which evaluated the accuracy of various image types and image processing approaches for mapping coastal ecosystem health indicators (Phinn et al., 2005, 2008).

Flight-lines were flown within 1 h of low tide, with a pixel size of 4×4 m and a swath of ~2.0 km. The flight-line plan minimized sun glint and hotspots by flying in and out of the solar plane (Dekker et al., 2001). The CASI-2 band-set was programmed to have 30 near-contiguous spectral bands in the visible-near infrared region (441–847 nm). The band selection of the CASI was designed to optimize the signal-to-noise ratio (SNR) across the spectrum by using variable bandwidths. The bands in the blue and near infrared region, where the sensor has a lower sensitivity, have a bandwidth of ~20 nm, while the bands in the 500–680 nm range have a ~10 nm bandwidth (Fig. 3). Due to the nature of the spectral band configuration, the full width at half maximum (FWHM) is equivalent to the bandwidth as it results from a combination of narrow channels with 1.3 nm FWHM.

2.2.2. Atmospheric and air water interface correction

As the first step of the integrated physics based mapping approach (Fig. 1), the 'coastal Waters and Ocean MODTRAN-4 Based ATmospheric correction' ('c-WOMBAT-c') procedure (Brando & Dekker, 2003; Phinn et al., 2005) was applied to achieve hyperspectral atmospheric correction of the CASI-2 imagery. The procedure combines an atmospheric inversion from at-sensor-radiance to above water reflectance (Adler-Golden et al., 1998; De Haan et al., 1997) with an inversion of the air–water interface from above water reflectance to subsurface reflectance (De Haan & Kokke, 1996; De Haan et al., 1997).

c-WOMBAT-c applies a full MODTRAN-4 atmosphere parameterization and characterisation to retrieve the subsurface remote-sensing reflectance (r_{rs}, [sr^{-1}]) from the CASI-2 at-sensor-radiance. The atmospheric parameterization for each flight-line was based on radiosonde data from the Australian Bureau of Meteorology Station at Brisbane

International Airport to estimate the atmospheric column water contents, the actual and the 24 h average wind speed. The estimate of ozone content was downloaded for the dates of CASI overflights from the Total Ozone Mapping Spectrometer—TOMS database (http://toms.gsfc.nasa.gov/ozone/ozone.html). The MODTRAN-4 summer mid-latitude atmosphere with navy maritime aerosol model and a horizontal visibility of 50 km was used, as this most closely matched the conditions at the time the image was taken.

In c-WOMBAT-c, adjacency effects from photons transferring from adjacent pixels to the one being sampled are corrected for by using an averaged surface radiance for the surrounding region. This spatially weighted image is generated by convolving the input radiance imagery

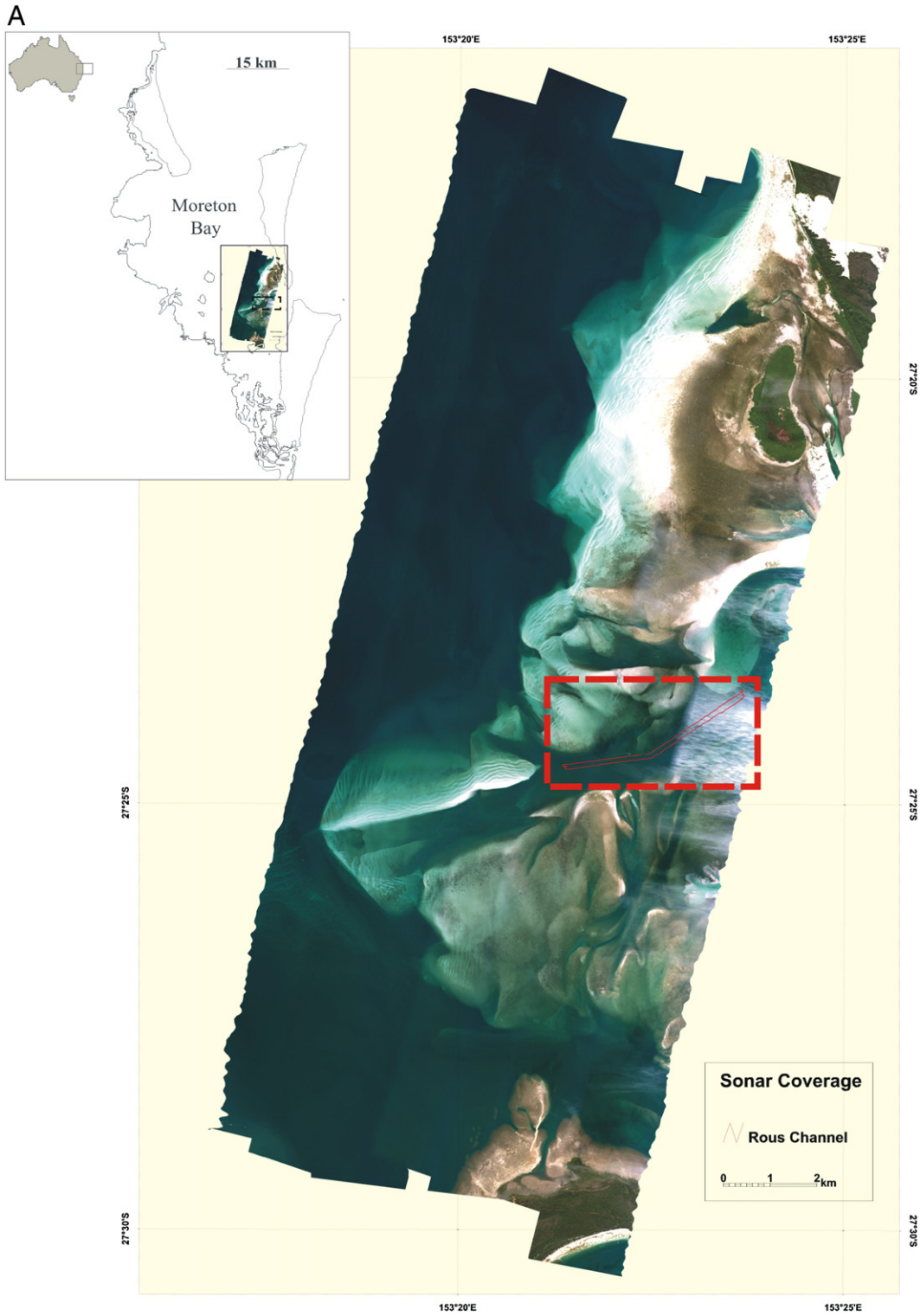

Fig. 2. Study area. A) Pseudo true colour composite of the CASI-2 imagery for the western portion of the Eastern Banks. The map is presented with the outlines of the bathymetric surveys. B) Pseudo true colour composite of the CASI-2 imagery for the Rous Channel area with the extent of the acoustic bathymetric survey overlaid as the orange thick polygon. The image is presented with the bathymetry vectors supplied by the Queensland Department of Transport.

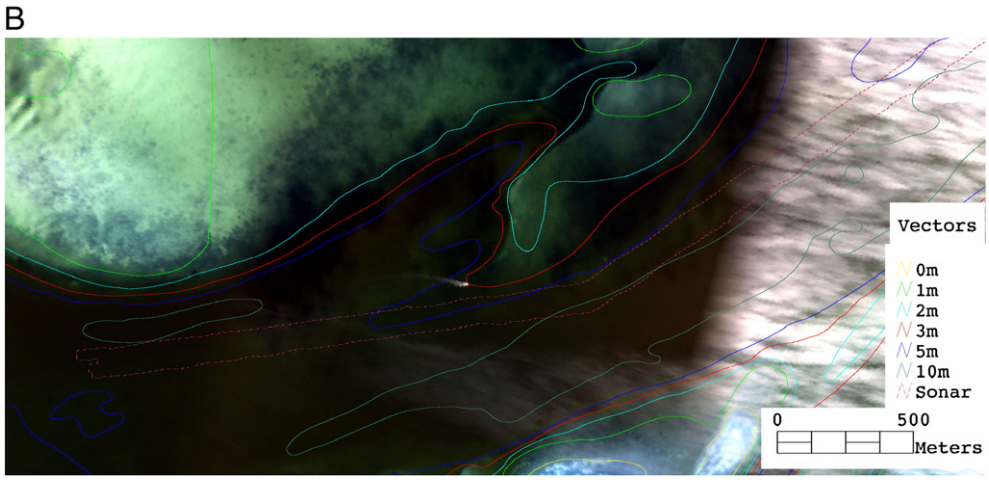

Fig. 2 (*continued*).

with a 1 km^2 spatial weighting function (Adler-Golden et al., 1998). The code is available from the authors upon request.

2.2.3. Illumination and geometric correction and cloud masking

After atmospheric correction, a residual brightness variation across each track-line could still be observed in several flight-lines. Cross track illumination variation is due to several physical environmental and sensor effects such as varying atmospheric path length, Fresnel reflection of diffuse skylight, camera lens and diaphragm effects, vignetting effects, instrument scanning, CCD illumination effects, internal scattering in the sensor and other non-uniform illumination effects. A cross-track correction for illumination variation was applied to remove these distortions. For each flight-line, a Row and Column Analysis of Variance (RC-ANOVA) was computed to correct for the systematic across-track illumination variation. The means and variances of the columns were used to balance each image band to the scene statistics (Brando & Dekker, 2003; Datt et al., 2003). In a scene that covers land, shallow water and deep water targets, scene statistics that are calculated over the entire image would not be representative for the deeper water targets. Therefore, an "as homogeneous as possible" water area of 50–100 lines was used as input for the statistics of each flight-line. For each spectral band, a 3rd order polynomial was fitted to the average value of each column after which an additive column-by-column RC-ANOVA correction was performed on the imagery.

The CASI-2 data were geometrically corrected and geo-referenced by the data provider, using data from the aircraft's inertial measurement unit and an onboard GPS unit referenced to a GPS base station. Following this, a manual adjustment and matching of each flight line to Landsat 5 Thematic Mapper image collected in August 2004 was performed to minimize the residual error in geo-location (Phinn et al., 2006, 2008).

A preliminary masking of the clouds based on their spectral properties was attempted but it was not adopted as it showed a low accuracy: very shallow portions of the Eastern Banks are very bright with reflectance properties similar to those of clouds in most spectral bands (see Fig. 2B). Thin and thick cloud masking was performed later as part of the quality control process of the integrated physics based mapping approach (see Sections 2.4.6 and 3.1).

2.2.4. Environmental dynamic range

In order to understand the precision and accuracy that can be achieved in the estimate of an environmental variable derived from reflectance with hyperspectral imagery, it is necessary to estimate the overall sensitivity of the entire sensor–atmosphere–air–water interface system for detecting changes in reflectance. The environmental

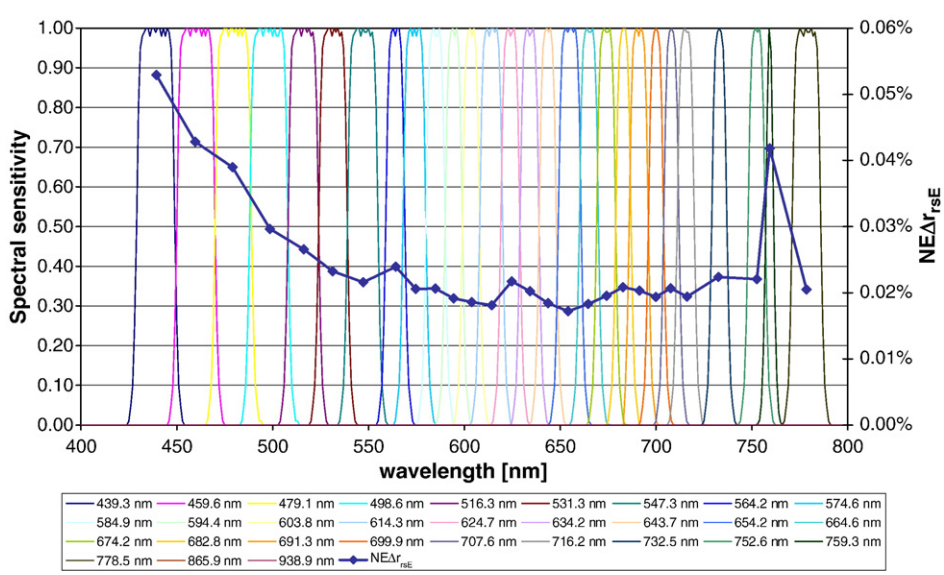

Fig. 3. CASI-2 programmable band set: spectral sensitivity (left axis) and noise equivalent difference in reflectance $NEDr_{rsE}$ (right axis).

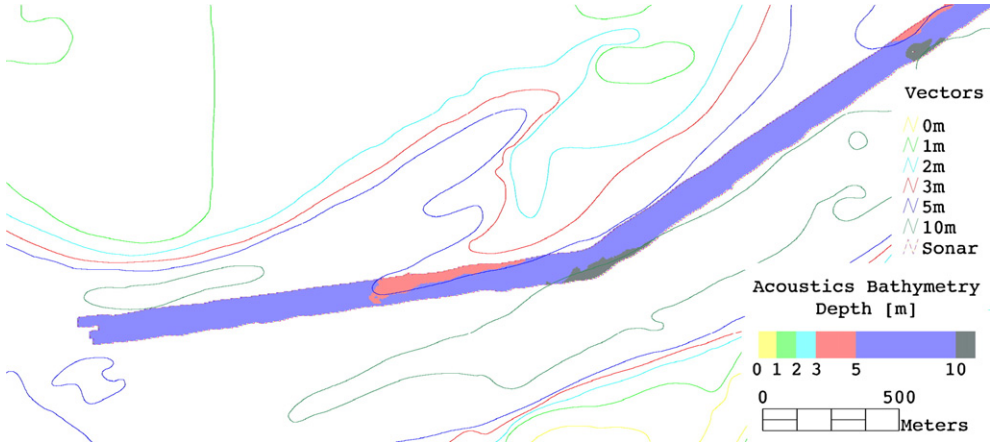

Fig. 4. Bathymetric surface derived from the acoustic bathymetric survey for the Rous Channel area. The bathymetry map is presented with the bathymetry vectors supplied by the Queensland Department of Transport.

noise equivalent reflectance difference (NEΔr_{rsE}) provides an integrated measure of sensor signal-to-noise ratio and scene-specific characteristics (Brando & Dekker, 2003; Wettle et al., 2004). For aquatic remote sensing, examples of the latter include atmospheric variability, effects from the air–water interface such as swell, wave and wavelet induced reflections, and refractions of diffuse and direct sunlight (Brando & Dekker, 2003).

The second step of the integrated physics based mapping approach (Fig. 1) is to estimate NEΔr_{rsE} in the deepest waters in the CASI-2 mosaic in the location identified as being the most homogenous using the methodology described by Wettle et al., (2004). Fig. 3 shows that NEΔr_{rsE} is lower than 0.00025 [sr^{-1}] in the 500–800 nm spectral range, with the exception of the very narrow band centred at 760 nm. As distinguishable levels of 0.00025 [sr^{-1}] are desirable (Brando & Dekker, 2003; Dekker et al., 2001), the quality of this imagery was deemed appropriate for further analysis.

2.3. Acoustic bathymetry data

As part of the larger project which evaluated the accuracy of various forms or image types and image processing approaches for mapping coastal ecosystem health indicators (Phinn et al., 2005, 2008), a number of underwater acoustical and video sensors were deployed in Moreton Bay between 29 August and 5 September 2004 (Siwabessy et al., 2006). The field equipment used in the survey included a RESON8125 multi-beam, a SIMRAD EQ60 single beam and Klein 5500 sidescan sonar systems, as well as an underwater video system (Siwabessy, 2005). The objective was to establish the domains, in terms of depths and water clarity, under which each data-set does and does not function effectively.

For the bathymetric survey, the RESON 8125 multi-beam survey lines were run parallel to the contour in order to get the maximum swath coverage in a short time. Accurate navigation and correction for motion allowed for stitching of overlapping swath lines and suppression of motion artefacts (Siwabessy, 2005). Raw bathymetry data were corrected for roll, pitch, yaw and GPS latency, as well as for the tide and the refraction. The corrected bathymetry was edited by removing bad data points/spikes. The bathymetry data were normalized to the Australian Height Datum and then re-sampled to the pixel size of the CASI imagery.

The uncertainty in the depth soundings was likely to be no more than a few centimetres, as the errors relate to the tidal model used to process the data, the motion sensor errors and the positional accuracy. These effects were corrected for whilst processing the bathymetry (Siwabessy, 2005).

Although approximately 82% of the Eastern Banks area is 3.0 m or shallower in depth (Phinn et al., 2005, 2008), the acoustic bathymetric survey for the Rous Channel area was carried out in a channel 5–10 m deep with a sandy substratum. Fig. 4 presents the bathymetric surface derived from the acoustic bathymetric survey for the Rous Channel area. The bathymetric surface corresponds to and accurately aligns with the rather coarse bathymetry vectors (1, 2, 3, 5, 10 and 20 m isobaths) which were based on a compilation of older data and were supplied by the Queensland Department of Transport. The acoustic

Table 1
Symbols and definitions

Symbol	Description	Units
CDOM	Coloured dissolved organic matter	
NAP	Non algal particles	
C_{CHL}	Concentration of chlorophyll-a	$\mu g \cdot L^{-1}$
C_{CDOM}	Measure of coloured dissolved organic matter	m^{-1}
C_{NAP}	Concentration of non algal particles	$Mg\ L^{-1}$
a	Total absorption coefficient, $a_w+a_{phy}+a_{CDOM}+a_{NAP}$	
a_w	Absorption coefficient of pure seawater	m^{-1}
a_{phy}	Absorption coefficient of phytoplankton pigments	m^{-1}
a_{CDOM}	Absorption coefficient of CDOM	m^{-1}
a_{NAP}	Absorption coefficient of NAP	m^{-1}
b_b	Total backscattering coefficient, $b_{bw}+b_{bp}$	m^{-1}
b_{bw}	Backscattering coefficient of pure seawater	m^{-1}
b_{bp}	Backscattering coefficient of suspended particles, $b_{bphy}+b_{bNAP}$	m^{-1}
b_{bphy}	Backscattering coefficient of phytoplankton particles	m^{-1}
b_{bNAP}	Backscattering coefficient of NAP	m^{-1}
$a^*_{phy}(\lambda)$	Chlorophyll-a specific absorption spectrum	m^2mg^{-1}
S_{CDOM}	Spectral slope constant for CDOM absorption coefficient	nm^{-1}
$a^*_{NAP}(\lambda_0)$	Specific absorption of NAP at the reference wavelength	m^2g^{-1}
S_{NAP}	Spectral slope constant for NAP absorption coefficient	nm^{-1}
λ_0	Reference wavelength	nm
$b^*_{bphy}(\lambda_0)$	Specific backscattering of algal particles at the reference wavelength	m^2mg^{-1}
Y_{phy}	Power law exponent for the algal particles backscattering coefficient	–
$b^*_{bNAP}(\lambda_0)$	Specific backscattering of NAP at the reference wavelength	m^2g^{-1}
Y_{NAP}	Power law exponent for NAP backscattering coefficient	–
r_{rs}	Subsurface remote-sensing reflectance for an optical shallow water body	sr^{-1}
r_{rs}^{dp}	Subsurface remote-sensing reflectance over a hypothetical optically deep water column	sr^{-1}
r_{rs}^{input}	Measured subsurface remote-sensing reflectance	sr^{-1}
r_{rs}^{model}	Modelled subsurface remote-sensing reflectance	sr^{-1}
$r_{rs}^{dp_model}$	Optically deep component of r_{rs}^{model}	sr^{-1}
NEΔr_{rsE}	Noise equivalent difference in reflectance	sr^{-1}
A	Substratum albedo (irradiance reflectance)	–
K_d	Vertical attenuation coefficient for diffuse downwelling light	m^{-1}
κ_B	Vertical attenuation coefficient for diffuse upwelling light originating from the bottom	m^{-1}
κ_C	Vertical attenuation coefficient for diffuse upwelling light originating from each layer in the water column.	m^{-1}
Δ	Optimization residuum	–
SD	Spectral substratum detectability	sr^{-1}
SDI	Substratum detectability index	–

bathymetric survey (Fig. 4) will be used for validation of the bathymetry retrieved with the integrated physics based mapping approach applied in this study to the hyperspectral imagery.

2.4. Retrieval of bathymetry, substratum composition and the optically active constituents

The third and pivotal step of the integrated physics based mapping approach (Fig. 1) is a physics based retrieval of bathymetry, substratum composition (i.e. fractional cover of e.g. sand, silt, mud, seagrass and macroalgae) and the concentrations of the optically active constituents of the water column (chlorophyll, CDOM and NAP) from the r_{rs} imagery. To this aim, the inversion/optimization method by Lee et al. (1999, 2001, 1998) was enhanced in order to: 1) retrieve the concentrations of optically active constituents in the water column (chlorophyll-a, CDOM and NAP), 2) account for more than one substratum cover type and 3) to estimate the contribution of the substratum to the remote sensing signal. This implementation, called SAMBUCA (the semi-analytical model for bathymetry, un-mixing, and concentration assessment), is available from the authors upon request.

2.4.1. Principles of the physics based method

At the core of the inversion/optimization method by Lee et al. (1999, 2001, 1998) lies an analytical expression for r_{rs} for an optical shallow water body (Maritorena et al., 1994):

$$r_{rs} = r_{rs}^{dp} + \exp(-K_d H)\left[A\exp(-\kappa_B H) - r_{rs}^{dp}\exp(-\kappa_C H)\right] \quad (1)$$

where, r_{rs}^{dp} is subsurface remote-sensing reflectance over a hypothetical optically deep water column; H is the water depth; A, the bottom albedo (substratum reflectance); K_d, the vertical attenuation coefficient for diffuse downwelling light, κ_B, the vertical attenuation coefficient for diffuse upwelling light originating from the bottom; and κ_C, the vertical attenuation coefficient for diffuse upwelling light originating from each layer in the water column (Table 1). Note that the attenuation of the upward flux is not equivalent to the attenuation of the downward flux (K_d). Attenuation of the upward flux must further be separated into a component originating from the water column (κ_C) and that originating from the bottom (κ_B) (Maritorena et al., 1994). By relating the four quantities r_{rs}^{dp}, K^d, κ_B and κ_C to absorption and backscattering via a series of semi-analytical relationships, Lee et al. (1999, 2001, 1998) modelled the r_{rs} spectrum as a function of five independent variables (representing properties of water column and bottom):

$$r_{rs} = f(P,G,X,B,H) \quad (2)$$

where P, G, X, and B are scalar values and represent absorption coefficients of phytoplankton and gelbstoff (coloured dissolved organic matter plus detritus), backscattering coefficient of suspended particles, and bottom reflectance at a reference wavelength, respectively; and H is the bottom depth.

2.4.2. Semi analytical model for bathymetry unmixing and concentration assessment (SAMBUCA)

In the inversion-optimization scheme in SAMBUCA the modelled subsurface remote-sensing reflectance (r_{rs}^{model}) is compared to the measured subsurface remote-sensing reflectance (r_{rs}^{input}) which was obtained from each pixel in the remote sensing image. The set of variables that minimizes the difference between these two spectra is used to estimate the environmental variables being sought, e.g. water column depth, substratum composition or the concentrations of the optically active constituents of the water column.

The extraction of environmental information from measured reflectance spectra constitutes a radiative transfer inverse problem. Inverse problems are notoriously difficult because of potential non-uniqueness issues (Mobley et al., 2005). It is often necessary to constrain inverse problems so as to guide the inversion to the correct solution. Such constraints often take the form of simplifying assumptions about the underlying physical or mathematical problem, or of added environmental information. For the inversion-optimization in SAMBUCA, the Downhill Simplex method was adopted, whilst ranges for variables to be optimized were constrained to reduce the occurrence of spectral ambiguities (Wettle et al., 2005; Wettle & Brando, 2006).

In SAMBUCA, the algorithm by Lee et al. (1999, 2001, 1998) was modified to retrieve the concentrations of optically active constituents in the water column (chlorophyll-a, CDOM and NAP). The absorption and backscattering coefficients are described as the sum of the contributions of N constituents and a constant coefficient for pure water:

$$a = a_w + \sum_{j=1}^{N} a_j^* C_j \,;\, b_b = b_{bw} + \sum_{j=1}^{N} b_{bj}^* C_j \quad (3)$$

where a_w and b_{bw} are the absorption and backscattering of pure water (Morel, 1974; Pope & Fry, 1997), a_j^*.and b_{bj}^* are the specific inherent optical properties (SIOPs) of jth constituent with concentration C_j. In the formulation of Eq. (3), CDOM has no backscattering term associated with it, and a_{CDOM} (440 nm) represents the concentration of CDOM.

The non-water absorption terms are parameterized as a known shape with an unknown magnitude:

$$a_{\text{phy}}(\lambda) = C_{\text{CHL}} \cdot a_{\text{phy}}^*(\lambda) \quad (4)$$

$$a_{\text{CDOM}}(\lambda) = C_{\text{CDOM}} \cdot a_{\text{CDOM}}^*(\lambda_0)\exp[-S_{CDOM}(\lambda-\lambda_0)] \quad (5)$$

$$a_{\text{NAP}}(\lambda) = C_{\text{NAP}} \cdot a_{\text{NAP}}^*(\lambda_0)\exp[-S_{NAP}(\lambda-\lambda_0)] \quad (6)$$

where C_{CHL} is the concentration of chlorophyll-a and $a_{\text{phy}}^*(\lambda)$ is the chlorophyll-a specific absorption spectrum. As the concentration of CDOM (C_{CDOM}) is represented by a_{CDOM} (440 nm), the reference wavelength λ_0 was set at 440 nm, S_{CDOM} is the spectral decay constant for CDOM absorption coefficient and $a_{\text{CDOM}}^*(\lambda_0)$ is set to 1. C_{NAP} is the concentration of NAP; $a_{\text{CDOM}}^*(\lambda_0)$ is the specific absorption of NAP at the reference wavelength, and S_{NAP} is the spectral slope constant for NAP absorption coefficient; and the reference wavelength λ_0 was set at 440 nm for NAP absorption coefficient.

The non-water backscattering terms are parameterized as follows:

$$b_{bp} = b_{b\text{phy}} + b_{b\text{NAP}} \quad (7)$$

$$b_{b\text{phy}}(\lambda) = C_{\text{CHL}} \cdot b_{b\text{phy}}^*(\lambda_0)\left(\frac{\lambda_0}{\lambda}\right)^{Y_{\text{phy}}} \quad (8)$$

$$b_{b\text{NAP}}(\lambda) = C_{\text{NAP}} \cdot b_{b\text{NAP}}^*(\lambda_0)\left(\frac{\lambda_0}{\lambda}\right)^{Y_{\text{NAP}}} \quad (9)$$

where $b_{b\text{phy}}^*(\lambda_0)$ is the specific backscattering of algal particles at the reference wavelength, Y_{phy}, the power law exponent for the algal particles coefficient; $b_{b\text{NAP}}^*(\lambda_0)$ is the specific backscattering of NAP at the reference wavelength, and Y_{NAP}, the power law exponent for NAP backscattering coefficient. The reference wavelength λ_0 was set at 542 nm for both algal and non algal particle backscattering coefficient.

In SAMBUCA, the algorithm by Lee et al. (1999, 2001, 1998) was modified to account for more than one substratum cover type in a pixel or spectrum by expressing the bottom albedo $A(\lambda)$ as a linear combination of two substrata:

$$A(\lambda) = q_{ij}A_i(\lambda) + (1-q_{ij})A(\lambda) \quad (10)$$

where q_{ij} represents the fractional cover of substratum i and substratum j within each pixel, $A_i(\lambda)$ and $A_j(\lambda)$ are the albedos of substratum i and j, respectively. When solving for more than two

cover types, SAMBUCA cycles through a given spectral library, retaining those two substrata and their estimated fractional cover q_{ij} which achieve the best spectral fit.

In summary, the complete model parameterization of Eq. (1) for SAMBUCA is:

$$r_{rs}^{\text{model}} = f\left(\begin{array}{l} C_{\text{CHL}}, C_{\text{CDOM}}, C_{\text{NAP}}, H, q_{ij}, A_i(\lambda), A_j(\lambda), S_{\text{CDOM}}, S_{\text{NAP}}, \\ Y_{\text{PHY}}, Y_{\text{NAP}}, a^*_{\text{PHY}}(\lambda), a^*_{\text{NAP}}(\lambda_0), b^*_{b\text{PHY}}(\lambda_0), b^*_{b\text{NAP}}(\lambda_0) \end{array}\right). \quad (11)$$

2.4.3. The parameterization of SAMBUCA

The parameterization of the semi-analytical model (Eq.(11)) relies on field sampling of the optical properties of the water body of interest. When this is not possible, the semi-analytical model (Eq. (11)) can be parameterized with appropriate values from the literature. For this study, inherent and apparent optical properties of Moreton Bay waters in winter were measured as part of a fieldwork campaign undertaken to coincide with image acquisition and other field surveys from 27th of July to the 3rd of August 2004. *In-situ* absorption, attenuation, backscattering, reflectance and vertical attenuation were measured at 20 locations in the bay (Phinn et al., 2006), and water samples were collected for measuring *in vivo* absorption of CDOM, chlorophyll and NAP, as well as the NAP and chlorophyll concentrations. All these measurements were carried out with the same instruments and laboratory methods described in Oubelkheir et al., (2006) and Phinn et al., (2006). Reflectance spectra of substratum, benthic algal species and epiphytic algal and invertebrate species were measured as in Roelfsema et al. (2006).

Based on these field data, SAMBUCA was configured to estimate the concentrations of optically active constituents in the water column (chlorophyll-a, CDOM and NAP), water column depth, and benthic substratum composition that produces the best fit between modelled and measured r_{rs}. These five environmental parameters are solved for on a pixel-by-pixel basis.

Table 2 reports the allowed optimization ranges for the chlorophyll, CDOM, and NAP concentrations, and the fixed values for seven scalar parameters (S_{CDOM}, S_{NAP}, a^*_{NAP}(440 nm), $b^*_{b\text{phy}}$(542 nm), $b^*_{b\text{NAP}}$(542 nm), Y_{phy} and Y_{NAP}) while the chlorophyll-a specific absorption spectrum $a^*_{\text{phy}}(\lambda)$ is presented in Fig. 5A. The values of the SIOP parameters for this study (Table 2, Fig. 5A) are similar to those reported previously for Moreton Bay and other Australian coastal waters (Brando & Dekker, 2003; Oubelkheir et al., 2006; Qin et al., 2007). Y_{phy} and Y_{NAP} have the same value as they were estimated from the *in situ* particulate backscattering measured using a HydroScat-6 according to Oubelkheir et al. (2006).

In the area of interest of this study, where the acoustic survey data were available for validation (namely Rous Channel, Fig. 2B), only two bare substrata were identified: brown mud and bright sand (Phinn et al., 2006; Phinn et al., 2008; Siwabessy, 2005). Hence, as the focus of this work was to estimate bathymetry from hyperspectral imagery, SAMBUCA was run with only two substrata (Fig. 5B) to expedite the inversion.

2.4.4. Measuring the "optical closure", or goodness-of-fit, between the modelled and actual image reflectance values

In the inversion-optimization scheme in SAMBUCA, r_{rs}^{model} is compared to r_{rs}^{input} using a goodness-of-fit or, error function. The set of variables that minimizes the difference between these two spectra is used to estimate the environmental variables being sought, e.g. water column depth, substratum composition or the concentrations of the optically active constituents of the water column.

The optimization residuum, Δ, is the measure of the difference between the measured and modelled spectra. In SAMBUCA Δ, can be estimated according to either a spectral magnitude matching function (e.g. Least squares in Albert & Gege, 2006; Lee et al., 1999, 2001; Mobley et al., 2005) or spectral shape matching function (e.g. Spectral Angle Mapper in Kruse et al., 1993) or as a hybrid formulation that combines

Table 2
SAMBUCA parameterization and optimization ranges

Parameter	Fixed value	Optimization range
C_{CHL}		0.7 (0.4–1.0)
C_{CDOM}		0.08 (0.04–0.11)
C_{NAP}		2.8 (1.0–3.3)
S_{CDOM}	0.0157	
S_{NAP}	0.0106	
a^*_{NAP}(440 nm)	0.0048	
a^*_{bphy}(542 nm)	0.00038	
Y_{phy}	0.681	
$b^*_{b\text{NAP}}$(542 nm)	0.0054	
Y_{NAP}	0.681	

the spectral matching and the least squares minimum to balance the requirements for spectral shape and magnitude matching:

$$\Delta = \alpha * LSQ \quad (12)$$

where, $\alpha[sr^{-1}]$ is the spectral angle between reference spectra and the spectra of the pixel in question as defined in the Spectral Angle Mapper (SAM, Kruse et al., 1993) by:

$$\alpha = \cos^{-1} \frac{\sum_{i=1}^{N} w(\lambda i) * \left(r_{rs}^{\text{model}}(\lambda_i) * r_{rs}^{\text{input}}(\lambda_i)\right)}{\left[\sum_{i=1}^{N} \left(w(\lambda i) * r_{rs}^{\text{model}}\right)^2\right]^{1/2} \left[\sum_{i=1}^{N} \left(w(\lambda_i) * r_{rs}^{\text{input}}(\lambda_i)\right)^2\right]^{1/2}} \quad (13)$$

and *LSQ* is the Least Square Distance:

$$\text{LSQ} = \frac{\left[\sum_{i=1}^{N} w(\lambda_i) * \left[r_{rs}^{\text{model}}(\lambda_i) - r_{rs}^{input}(\lambda_i)\right]^2\right]^{1/2}}{\sum_{i=1}^{N} w(\lambda_i) * r_{rs}^{\text{input}}(\lambda_i)}. \quad (14)$$

The weighting function, $w(\lambda)$, can be introduced for both *LSQ* and α to weigh the contribution of different wavelength bands.

One of the primary advantages of α is that spectral angle is insensitive to differences in albedo, i.e. the magnitude of the measured and modelled spectrum. The spectral angle only measures differences in spectral shape, and spectral magnitude is measured by the length of each vector (Dennison et al., 2004; Kruse et al., 1993). Since the residual error is partially dependent on the reflectance of each band within the modelled spectrum, *LSQ* will be partially dependent on the magnitude of the modelled reflectance. As the magnitude of the modelled spectrum increases, *LSQ* will also increase (Dennison et al., 2004).

To assess the spectral match for the inversion, the hybrid formulation (Eq.(12)) for Δ was adopted, as it combines the strengths of spectral angle mapping with the strengths of a minimum distance classifier, ensuring that the resultant spectrum r_{rs}^{model} has the appropriate magnitude and shape as the input spectrum r_{rs}^{input}. In this study, $w(\lambda)$ was set to $1/\text{NE}\Delta r_{rsE}$ to discount the wavelengths where the signal is less accurate or noisier.

As a result of SAMBUCA's inversion-optimization scheme, Δ can be used as a measure of the goodness-of-fit, between the measured and modelled spectra (i.e. the optical closure). In other words, each set of retrieved environmental variables is assigned a confidence rating based on SAMBUCA's ability to model a given subsurface reflectance spectra.

2.4.5. Contribution of the substratum to the remote sensing signal–substratum detectability index

In SAMBUCA, the algorithm by Lee et al. (1999, 2001, 1998) has been modified to estimate the contribution of the substratum to the subsurface remote-sensing reflectance signal by comparing the modelled

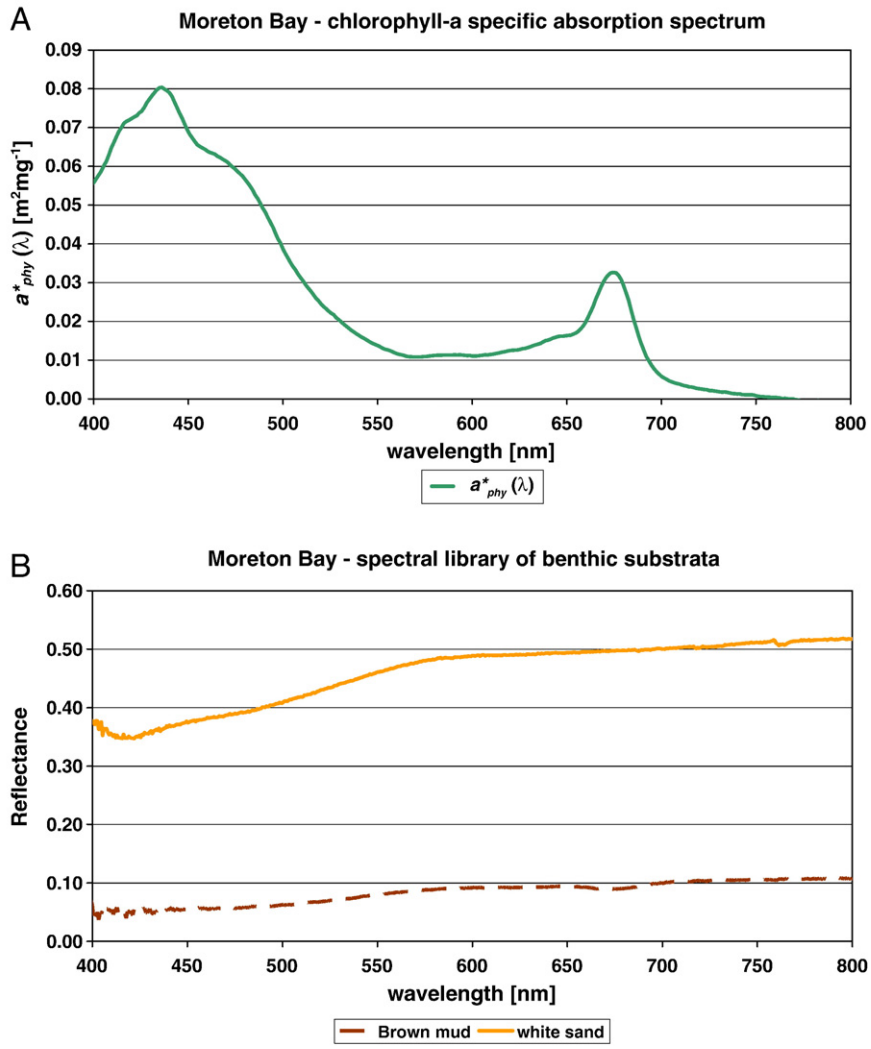

Fig. 5. Spectral parameterization for SAMBUCA. A) Chlorophyll-a specific absorption spectrum $a^*_{\mathrm{phy}}(\lambda)$. B) SAMBUCA's Spectral library of benthic substrata.

spectrum using an optically deep system ($r_{rs}^{dp_model}$, i.e. the term indicated as r_{rs}^{dp} in Eq. (1)), and the modelled spectrum for an optical shallow water body as generated by SAMBUCA (r_{rs}^{model}). Fig. 6 provides a graphical illustration of the contribution of the substratum to the remote sensing signal for two pixels extracted from the imagery by comparing $r_{rs}^{dp_model}$ and r_{rs}^{model}. Fig. 6A presents an example of optically shallow waters where the signal from the substratum is directly measurable, while Fig. 6B presents an example where the signal from the substratum is small and diffuse and close to the noise levels in the imagery.

To provide a quantitative indication of the contribution of the substratum to the subsurface remote-sensing reflectance signal of the water body, we introduce SD(λ), the spectral substratum detectability:

$$\mathrm{SD}(\lambda) = \left(r_{rs}^{\mathrm{model}} - r_{rs}^{dp_model}\right)/\mathrm{NE}\Delta r_{rsE}. \tag{15}$$

This quantity is sensor dependent and scene dependent: it quantifies the contribution of the substratum to the subsurface remote-sensing reflectance signal for a given sensor as it uses the noise equivalent difference in reflectance ($\mathrm{NE}\Delta r_{rsE}$) as a scaling factor. As can be shown by rearranging Eq. (1):

$$\mathrm{SD} = \left(\exp(-K_d H)\left[A\exp(-\kappa_B H) - r_{rs}^{dp_model}\exp(-\kappa_C H)\right]\right)/\mathrm{NE}\Delta r_{rsE}. \tag{16}$$

The SD(λ) is function of the vertical attenuation coefficients (K_d,K_g and K_c), noise equivalents.

Furthermore, we introduce the substratum detectability index (SDI) defined as the absolute value of the spectral substratum detectability for the band of maximum penetration:

$$\mathrm{SDI} = \max\left(\left|r_{rs}^{\mathrm{model}} - r_{rs}^{dp_model}\right| / \mathrm{NE}\Delta r_{rsE}\right) \tag{17}$$

Fig. 6(C, D) illustrates the derivation of the spectral substratum detectability (SD(λ)) and the substratum detectability index (SDI) from the two modelled spectra resulting from the inversion optimization (r_{rs}^{model} and $r_{rs}^{dp_model}$ in Fig. 6(A, B)). For the optically shallow water pixel (Fig. 6C) the signal from the substratum $|r_{rs}^{\mathrm{model}} - r_{rs}^{dp_model}|$ is larger than 0.0005 in the 500–600 nm spectral range, and SD(λ) is higher than 5 in the same range, with a maximum of 15 at ~580 nm (i.e. SDI = 15). In Fig. 6D, the signal from the substratum is small and diffuse and $|r_{rs}^{\mathrm{model}} - r_{rs}^{dp_model}|$ is larger than 0.0005 only at 500 nm, SD(λ) is therefore equal to 1 or 2 in the 480–580 spectral range, with a maximum of 2 at ~500 nm (i.e. SDI = 2).SDI allows three classes of waters to be identified in the imagery:

- "optically shallow waters" where the signal from the substratum is directly measurable and the substratum signal at the surface is more than 5 $\mathrm{NE}\Delta r_{rsE}$ at the band of maximum penetration (SDI > ~5, e.g. Fig. 6(A, C));
- "quasi-optically deep waters", where the contribution from the substratum is weak and the substratum signal at the surface is between 1 and 5 $\mathrm{NE}\Delta r_{rsE}$ (i.e. 1 < SDI < 5, e.g. Fig. 6(B, D)); and

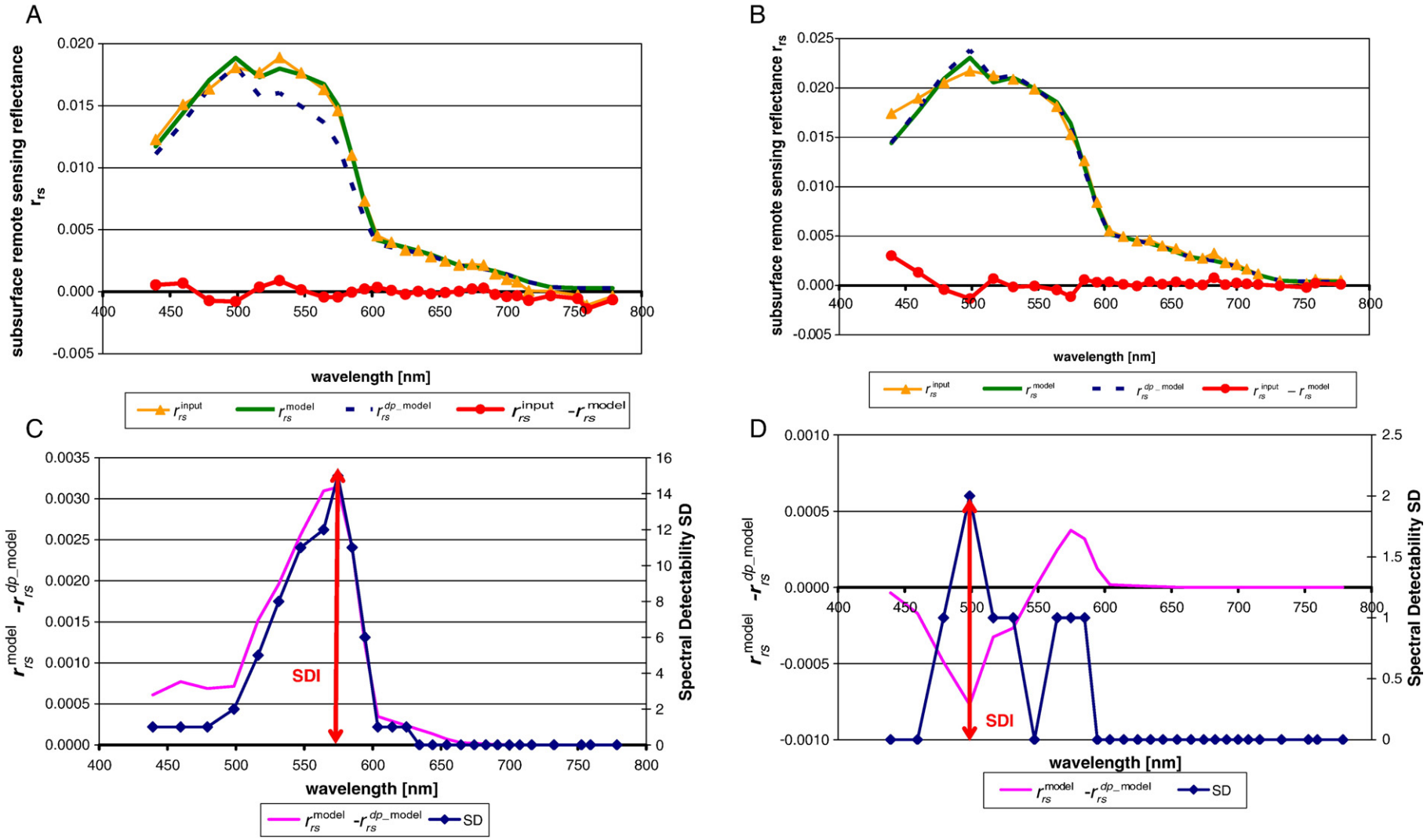

Fig. 6. Graphical representation of SAMBUCA's main spectral quantities for two pixels extracted from the imagery. r_{rs}^{input} is the measured remote-sensing irradiance reflectance; r_{rs}^{model} is the modelled subsurface remote-sensing reflectance resulting from the inversion optimization of r_{rs}^{input}; $r_{rs}^{dp_model}$ is the optically deep component of r_{rs}^{model}; SD is spectral substratum detectability. a and c present an example of "optically shallow waters" where the signal from the substratum is directly measurable (H=6.9 m, SDI=15); while b and d present an example of "quasi optically deep waters" where the signal from the substratum is small and diffuse (H=10.7 m, SDI=2).

- "optically deep waters", where no signal from the substratum is measurable (i.e. SDI = 0).

For optically deep waters (i.e. SDI = 0) the estimate of depth is not possible as no signal from the substratum is measurable. Because of that, the retrieved values of the water depth from the inversion/optimization can be any of the values deeper than the depth that sets SDI to 0, leading to an over-estimate of the real depth for those pixels. To retrieve an "as shallow as possible" depth when an optically deep solution is encountered at the end of a SAMBUCA inversion optimization, a secondary iteration was introduced. The retrieved depth for the optically deep pixels is iteratively decreased while maintaining SDI = 0. This secondary iteration ensures the retrieval of an "as shallow as possible" depth value for each pixel.

It is worth mentioning that both SD(λ) and SDI are sensor- and scene-dependent and they could be used to compare the contribution of the substratum to the remote sensing signal for different sensors or imagery acquired in different environmental conditions.

2.4.6. Integrating optical closure and SDI to exclude inappropriate data

The fourth step of the integrated physics based mapping approach (Fig. 1) is an objective process of quality control of the raw SAMBUCA output. This is based on the combined analysis of two ancillary variables of the SAMBUCA output, the measure of optical closure (i.e. the

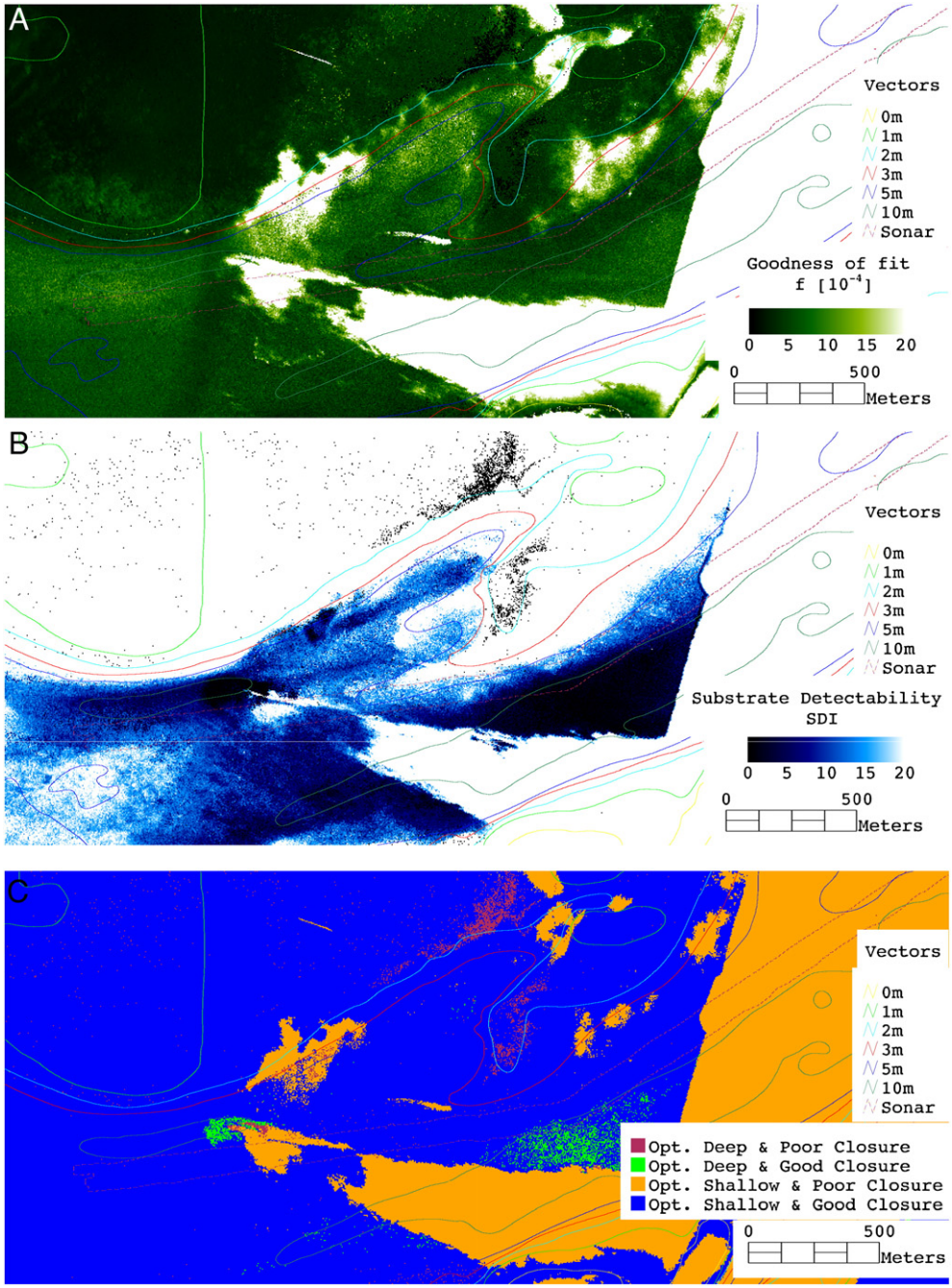

Fig. 7. Process of quality control of the SAMBUCA depth retrieval from CASI-2 image for the Rous Channel area. A) Map of the optimization residuum (D, lower values indicate a good fit). B) Map of substratum detectability index (higher values indicate a higher signal from the substratum). C) Guide to the interpretation of the SAMBUCA inversion of CASI-2 image for the Rous Channel area. The maps are overlaid with the extent of the acoustic bathymetric survey and the bathymetry vectors supplied by the Queensland Department of Transport.

Table 3
Process of quality control of the SAMBUCA depth retrieval from CASI-2 image for the Rous Channel area

	Poor optical closure (i.e. $\Delta>0.002$)	Good optical closure (i.e. $\Delta<0.002$)
Optically deep (i.e. with SDI=0)	Pixels occurring in few small patches in the northern portion of the imagery (they could be cloud shadows)	Pixels occurring at the knee-bend in sonar coverage and around the patch of pixels with a poor closure and optically deep in the western portion of the imagery
Optically shallow (i.e. with SDI>1)	Pixels occurring in correspondence with the thick clouds in the eastern portion of the imagery	Pixels corresponding to the overlap of the pixels mapped out in brighter tones in Fig. 7A and B

Combination of the optimization residuum and the substratum detectability index (SDI).

optimization residuum, Δ) and the map of substratum detectability index (SDI). By defining a threshold of the optimization residuum, Δ_{thresh}, it is possible to identify the pixels with a good or poor spectral match for the inversion, while the substratum detectability index (SDI) indicates whether the pixels are optically deep or optically shallow. The combined use of these additional outputs enables the identification of four classes of pixels:

- pixels with a poor optical closure (i.e. $\Delta>\Delta_{thresh}$) and optically shallow or quasi-optically deep (i.e. with SDI>1);
- pixels with a poor closure and optically deep<i.e. with SDI=0);
- pixels with a good closure (i.e. $\Delta<\Delta_{\mathrm{thresh}}$) and optically deep;
- pixels with a good closure and optically shallow or quasi-optically deep.

Only the pixels belonging to the last class should be used for further analysis of the bathymetry retrieval, since the good optical closure ensures that a good spectral match for the inversion was attained, while SDI>1 ensures that the estimate of depth is possible as the signal from the substratum is measurable. This systematic quality control procedure facilitates the use of sub-optimal remote sensing imagery as it assesses objectively the data quality on a pixel-by-pixel basis.

3. Results and discussion

SAMBUCA was applied to the CASI-2 imagery for the Rous Channel for the retrieval of depth, concentrations of optically active constituents in the water column (chlorophyll-a, CDOM and NAP) and substratum composition. To demonstrate the effect of the physics based quality control procedure (Fig. 7 and Table 3), the bathymetry output will be shown before and after the procedure (Figs. 8 and 10) followed by the comparison of the retrieved depth with the acoustic depth (Figs. 9 and 11). The maps of concentrations of chlorophyll, CDOM and NAP and the substratum composition as retrieved by SAMBUCA will be not shown or discussed in this paper for the sake of brevity. The retrieved ranges of C_{CHL}, C_{CDOM}, and C_{NAP} were respectively 0.4–1.0 μg L^{-1}, 0.04–0.11 m^{-1} and 1.0–3.3 mg L^{-1}. The retrieved substratum albedo A(550 nm) ranged between 8 and 40%.

Fig. 8 presents the bathymetric surface derived from CASI-2 image for the Rous Channel area. This map is the raw SAMBUCA output as it presents the estimated bathymetry for all the pixels in the imagery before quality control. Overall, this bathymetric surface is accurately aligned with the bathymetry vectors supplied by the Queensland Department of Transport which were overlaid on all the maps for all of the isobaths. Furthermore, a visual comparison of the bathymetric surface derived from the CASI-2 image and the bathymetric surface derived from the acoustic bathymetric survey (Figs. 8 and 4 respectively) show a generally good agreement. The effects of the sub-optimal quality of the CASI-2 imagery on the depth retrieval can be seen in the eastern portion of the imagery which is affected by thick clouds. It is mapped as exposed ($Z<0$ m AHD; Australian Height Datum). Furthermore, thin clouds in the central portion of the imagery are mapped as very shallow ($Z<3$ m AHD).

3.1. Quality control procedures of the SAMBUCA output

The process of quality control of the raw SAMBUCA bathymetry output involves the combined analysis of two ancillary output variables: the measure of optical closure (i.e. the optimization residuum, Δ, Fig. 7A) and the map of substratum detectability index (SDI, Fig. 7B).

A visual analysis of the map of optical closure (Fig. 7A) identifies areas of the image where the presence of thick or thin clouds (as evident in Fig. 2B) creates a poor optical closure (i.e. $\Delta>\sim0.002$). These pixels have spectra that SAMBUCA cannot solve appropriately, leading to a poor optical closure. As stated in Section 2.2.3, a preliminary supervised masking of the clouds based on their spectral properties was attempted but it was not adopted as it showed a low accuracy. Based on the visual analysis of the map of the optimization residuum, Δ, the threshold of the optimization residuum, Δ_{thresh} was set to 0.002 in order to separate good and poor optical closure. Δ_{thresh} is sensor and scene dependent as Δ, incorporates $\mathrm{NE}\Delta r_{rsE}$ as weighting function to discount the wavelengths where the signal is less accurate or noisier.

The map of substratum detectability index (Fig. 7B) identifies the areas that are retrieved by the model as optically deep (i.e. where no effects of the substratum are measurable) and those where there are varying levels of bottom visibility.

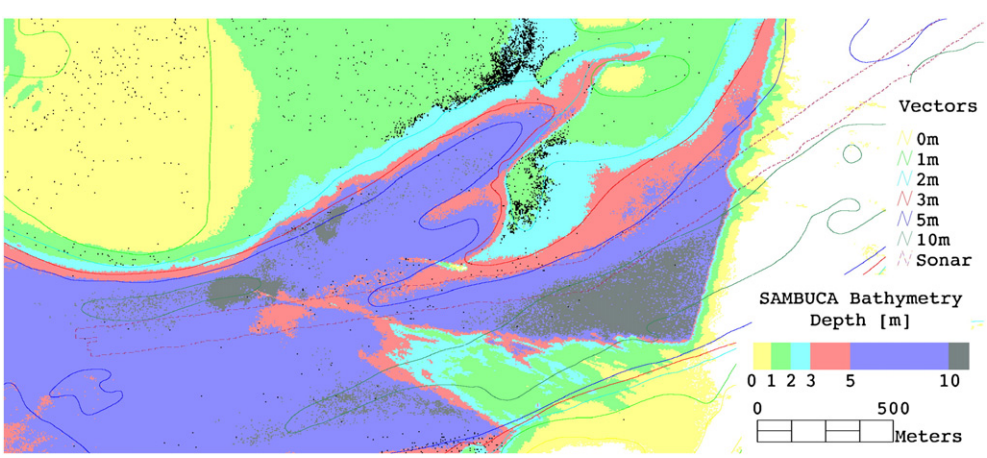

Fig. 8. Non quality controlled bathymetric surface derived from CASI-2 image for the Rous Channel area with the extent of the acoustic bathymetric survey overlaid as the purple polygon. The bathymetry map is presented with the bathymetry vectors supplied by the Queensland Department of Transport.

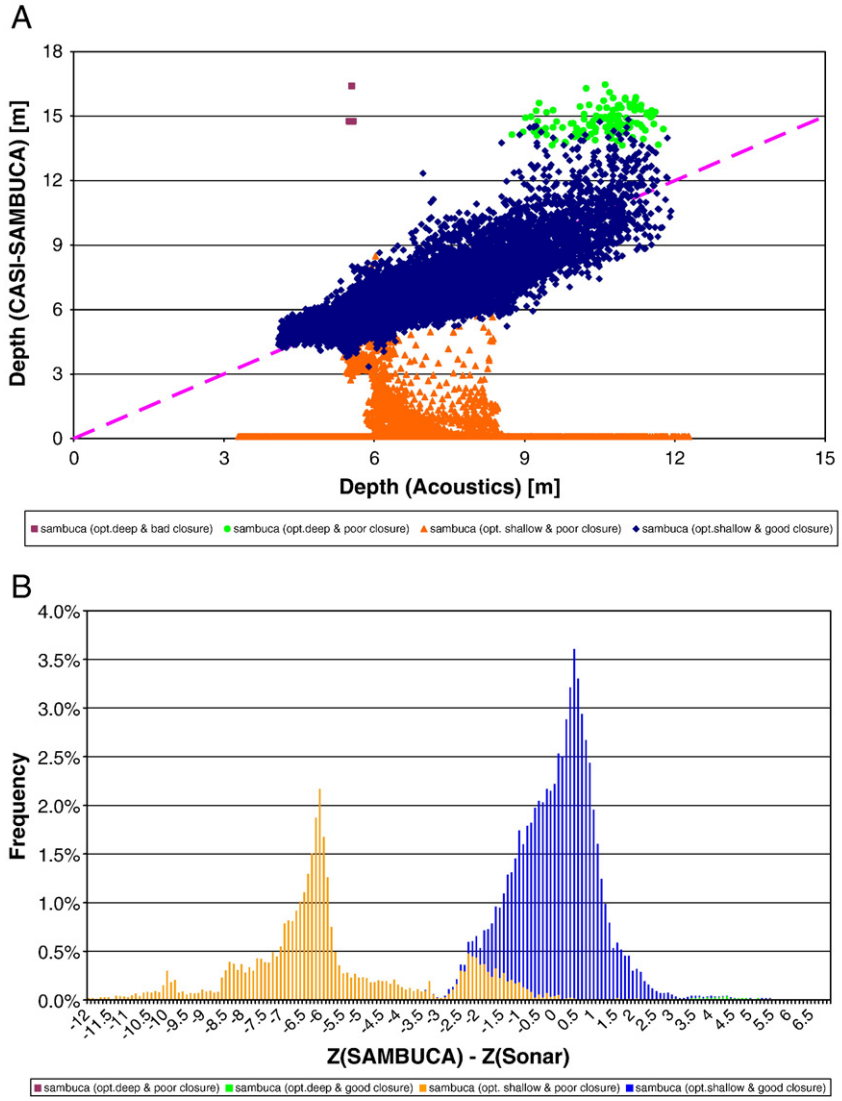

Fig. 9. Comparison of depth estimated from CASI-2 image with the depth estimated with the acoustic bathymetric survey for the Rous Channel area A) Scatterplot of the depth estimated from CASI-2 image versus the depth estimated with the acoustic bathymetric survey. B) Histogram of the error in the depth estimated from CASI-2 image. The points are colour coded according to the classes mapped in Fig. 7C.

By combining the analysis of the map of optical closure (Fig. 7A) and the map of index of optical depth (Fig. 7B), the four classes of pixels identified in Section 2.4.6, can be observed in Fig. 7C. Table 3 provides details of the pixels occurring in each of those classes in the CASI-2 scene. The pixels with a good closure (i.e. $\Delta<0.002$) and optically shallow (i.e. with SDI>1), correspond to the overlap of the pixels mapped out in brighter tones in Fig. 7A and B. Only these pixels should be used for further analysis of the bathymetry retrieval, since the good optical closure ensures that a good spectral match for the inversion was attained, SDI>1 ensures that the estimate of depth is possible as the signal from the substratum is measurable. The combined use of the additional output enabled the detection and masking of thin and thick clouds as well as cloud shadows, as these pixels have spectra that SAMBUCA cannot solve appropriately and hence it would lead to a poor optical closure.

3.2. Effects of the quality control on the SAMBUCA bathymetric output

Fig. 9A presents the scatter-plot of the depth estimated from CASI-2 image (before quality control) versus the depth estimated with the acoustic bathymetric survey for the Rous Channel area. Fig. 9B presents the distribution of the difference of the depth as estimated from CASI-2 image with the acoustic bathymetric survey for the Rous Channel area. For both charts the points are colour coded according to the four classes mapped in Fig. 7C. In Fig. 9 it is possible to identify the four classes of Fig. 7C and Table 3 as three separate clusters:

i) the optically shallow pixels with poor closure and, occurring in correspondence with the thick clouds in the eastern portion of the imagery are mapped as exposed (Z=0.1 m AHD) and the thin clouds in the central portion of the imagery, are mapped as very shallow (Z<3 m AHD);
ii) the optically deep pixels (SDI=0, with and without a good optical closure) are overestimating the depth by 3–10 m;
iii) the optically shallow pixels (i.e. for high values of SDI) with a good optical closure where the bathymetry estimates are centred around the 1:1 line.

Fig. 10 presents the quality controlled SAMBUCA output, where only the pixels with a good closure (i.e. $\Delta<0.002$) for optically shallow waters (i.e. with SDI> 1) are presented. A visual comparison of Fig. 10A and Fig. 4 show that most of the pixels in the SAMBUCA inversion are in the 5–10 m class with a smaller number of pixels in the 3–5 m class.

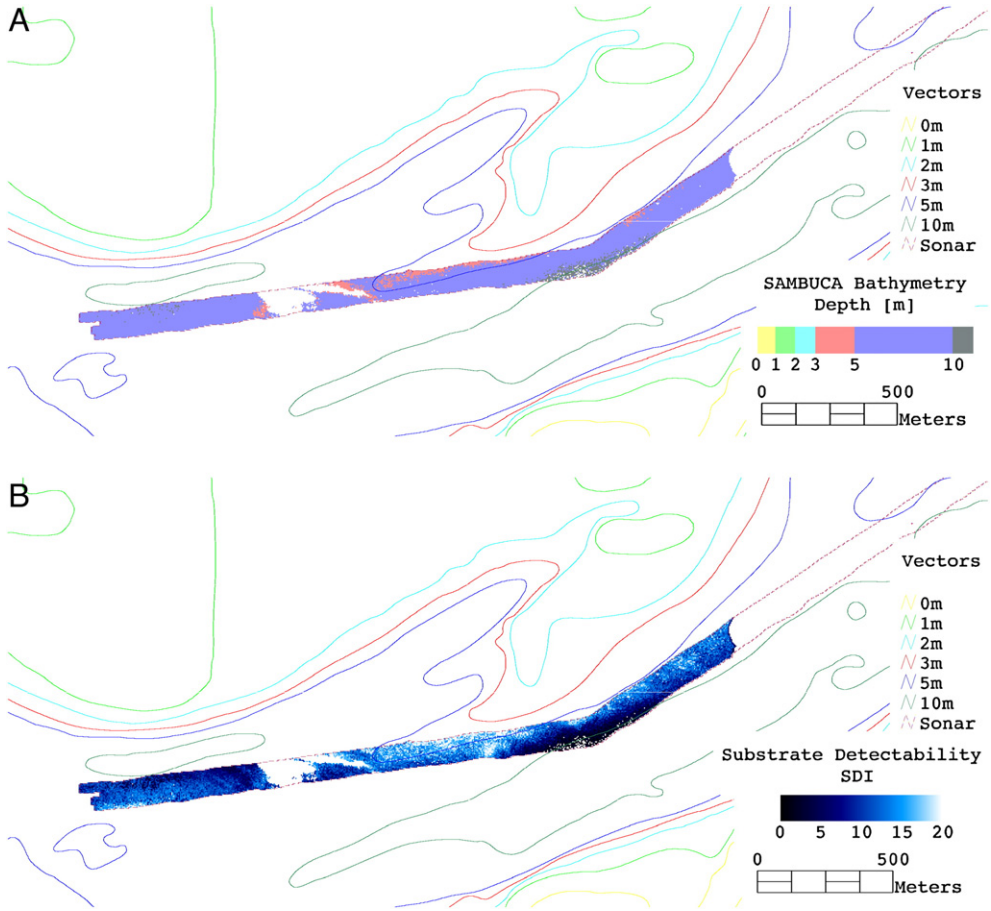

Fig. 10. Quality controlled results derived from CASI-2 image for the extent of the acoustic bathymetric survey for the Rous Channel area. A) Quality controlled bathymetric surface. B) Map of substratum detectability index (SDI). Only the pixels with a good closure (i.e. $\Delta < 0.002$) and not optically deep (i.e. SDI > 0) are represented. The maps are presented with the bathymetry vectors supplied by the Queensland Department of Transport.

These retrieved depths correspond to and accurately aligns with both the acoustic data and the bathymetry vectors supplied by the Queensland Department of Transport (Fig. 4).

This systematic quality control procedure facilitates the use of sub-optimal remote sensing imagery as it assesses objectively the data quality on a pixel-by-pixel basis.

3.3. Effects of the substratum contribution to the remote sensing signal on the precision and accuracy of the bathymetric retrieval

Since the substratum detectability index (SDI) provides an estimate of the contribution of the substratum to the sub-surface remote sensing signal, a higher SDI is a measure of increased substratum visibility. To investigate the effects of the optical depth on the precision and accuracy of the bathymetric retrieval using SAMBUCA and hyperspectral data, the comparison of the image-based depth retrieval to the acoustics-based depth retrieval is divided into four classes of SDI, representing an increase in bottom visibility as mapped in Fig. 10B.

Fig. 11A presents the scatter-plot of the depth estimated from CASI-2 image versus the depth estimated with the acoustic bathymetric survey for the Rous Channel area for four SDI classes. The points are colour coded according to the classes mapped in Fig. 10B. Fig. 11B presents the distribution of the difference between the depths as estimated from the CASI-2 images and the acoustic bathymetric survey for the Rous Channel area. Fig. 11 shows that the classes with lower SDI are able to map to deeper depth but with a lower level of precision. Conversely an increased substratum visibility (classes with a higher SDI) leads to a more precise depth retrieval. The "quasi-optically deep waters", i.e. the class with SDI values ranging from 1 to 5, show a tendency to overestimate the depth in similar fashion to the optically deep pixels in Fig. 9. For this class, there is some effect of the bottom visibility on the remote sensing signal, but a lower signal to noise ratio creates a low level of precision in the estimate of depth.

Three different statistics can be used to describe precision and accuracy of the depth retrieval:

$$\mathrm{Bias}\left(\hat{z},z\right) = \mathrm{mean}\left(\hat{z}\right) - \mathrm{mean}(z) \tag{18}$$

$$\mathrm{MedianDifference}\left(\hat{z},z\right) = \mathrm{median}\left(\hat{z}\right) - \mathrm{median}(z) \tag{19}$$

$$\mathrm{RMSE}\left(\hat{z}\right) = \sqrt{\mathrm{Var}\left(\hat{z}\right) + \left(\mathrm{Bias}\left(\hat{z},z\right)\right)^2} \tag{20}$$

where z is the true depth (i.e. the acoustic survey) and z is the depth estimated by SAMBUCA. *Bias* [m] and *MedianDifference*[m] provide an indication of accuracy in the measurement, while *RMSE* (Root Mean Square Error, [m]) describes the "overall accuracy" encompassing both random errors (i.e. affecting the precision of the measurement) and systematic errors (i.e. affecting the accuracy of the measurement).

Table 4 reports the precision and accuracy of the depth retrieval in the Rous Channel. As the comparison of Fig. 9 and Fig. 11 showed, *Bias*, *MedianDifference* and *RMSE* for all the pixels before the quality control procedure (−2.02 m, −0.63 and 3.85 m), decrease dramatically to 0.07 m, 0.16 m and 0.95 m for the pixels identified as having "good optical closure" and being optically shallow.

The bathymetric retrieval shows a *Bias* compared to the acoustic survey of 0.07 m for all the quality controlled pixels, a higher *Bias* of

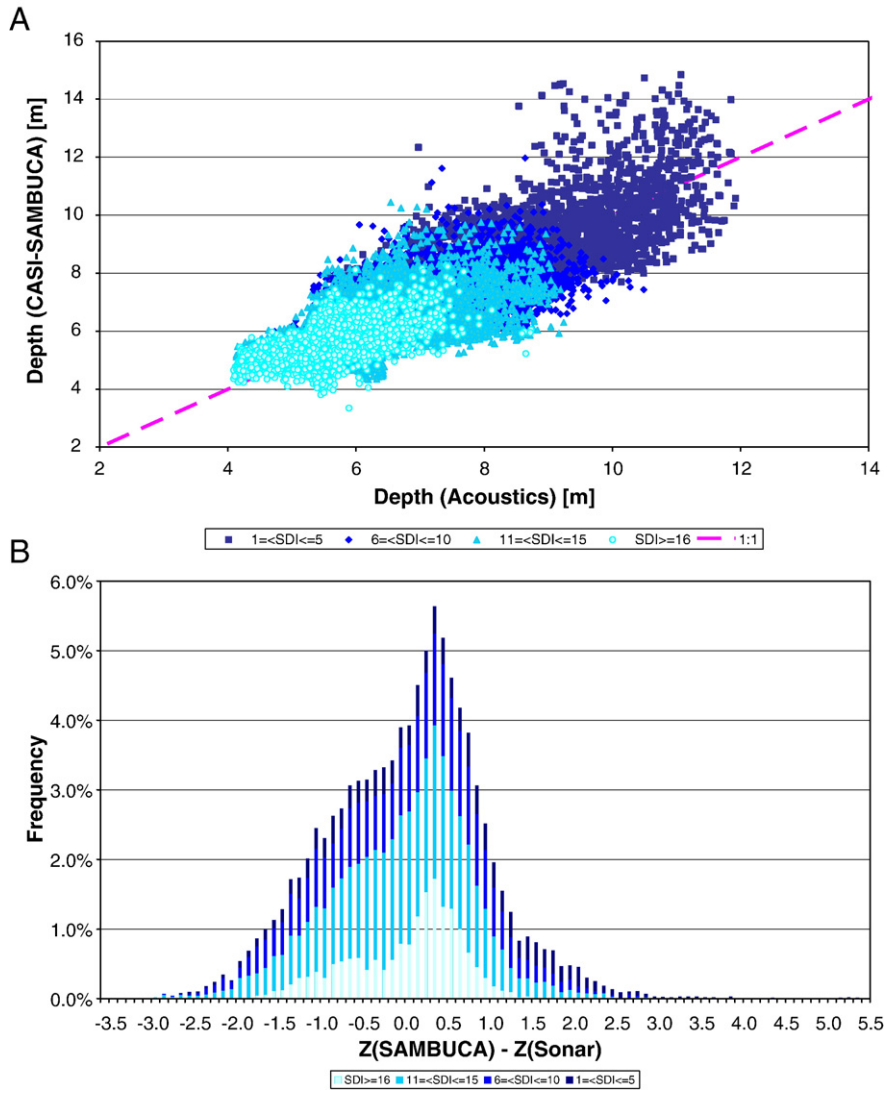

Fig. 11. Comparison of the quality controlled depth estimated from CASI-2 image versus the depth estimated with the acoustic bathymetric survey for the Rous Channel area for four substratum detectability index (SDI) classes. A) Scatter-plot of depth estimated from CASI-2 image versus the depth estimated with the acoustic bathymetric survey. B) Histogram of the error in the depth estimated from CASI-2 image for the Rous Channel area for the four SDI classes. Only the pixels with a good closure (i.e. $\Delta<0.002$) and not optically deep (i.e. with an SDI>0) are represented as in Fig. 10. The points are colour coded according to the classes mapped in Fig. 10B.

0.43 m for the quasi-optically deep waters, and a *Bias* ranging between −0.01 and −0.07 m for the remaining classes. An increased substratum visibility (classes with a higher SDI) lead to a lower *RMSE*, hence a more accurate retrieval (1.35 m or ±15% for 1<SDI<5 and up to 0.67 m or ±13% for SDI>15). This is controlled by the precision of the estimates, as both *Bias* and *MedianDifference* do not change significantly for the three shallow water classes. To our knowledge, the quantitative identification and screening of the "optically deep waters" and the "quasi-optically deep waters" was never attempted prior to this research, leading to a degraded accuracy in the depth retrievals for previous studies as the waters became deeper or more turbid (e.g. Goodman & Ustin, 2007; Lee et al., 2007; McIntyre et al., 2006; Sandidge & Holyer, 1998; Stumpf et al., 2003).

The precision of estimating bathymetry from remote sensing data is, as expected, a function of the contribution of the substratum to the subsurface remote-sensing reflectance signal, as quantified by SDI. For the "quasi optically deep waters" a lower signal to noise ratio creates a low level of precision in the estimate of depth. Most likely, the

Table 4
Statistics of the accuracy of the depth retrieval in the Rous Channel before and after the quality control procedure and for four classes of substratum detectability index (SDI)

	All points before quality control	All point after quality control	After quality control 1<SDI<5	After quality control 6<SDI<10	After quality control 11<SDI<15	After quality control 16<SDI<30
N	22706	14,450	1913	4084	5906	2547
Bias	−2.02	0.07	0.43	0.02	−0.01	0.07
Median difference	−0.63	0.16	0.42	0.14	0.05	0.25
RMSE	3.85	0.95	1.35	0.97	0.87	0.67
% error	–	14.2%	15.3%	14.2%	14.4%	12.9%

accuracy of the retrieval is a function of the calibration of the hyperspectral imagery, the parameterization of the atmospheric correction and the parameterization of the SAMBUCA model.

These results suggest that the integrated physics based mapping approach adopted in this study performs well for retrieving water column depths in coastal waters in water depths ranging 4–13 m for the area and conditions studied, even with sub-optimal imagery.

The ability of the physics-based quality control procedure in identifying only the pixels as having "good optical closure" and being optically shallow enabled a retrieval from suboptimal imagery with accuracy and precision comparable to other works carried out using optimal imagery (e.g. Adler-Golden et al., 2005; Klonowski et al., 2007; Lee et al., 2001; McIntyre et al., 2006).

4. Conclusion

In this study the inversion/optimization approach by Lee et al. (1999, 2001, 1998) was enhanced to retrieve the concentrations of optically active constituents in the water column (chlorophyll-a, CDOM and NAP), to account for more than one substratum cover type and to estimate the contribution of the substratum to the remote sensing signal.

The strength of applying the physics-based approach to airborne hyperspectral data was emphasised by its ability to produce outputs that can be used for quality control procedures. Despite inadequate quality of portions of the CASI-2 imagery due cloud cover in the area, the quality control procedure was able to identify pixels with a reliable retrieval of depth and to detect thin and thick clouds, as well as cloud shadows, which were all masked out from further analysis. This systematic quality control procedure facilitates the use of sub-optimal remote sensing imagery as it objectively assesses the data quality on a pixel-by-pixel basis. As optimal remote sensing acquisition conditions seldom occur, the adoption of this systematic quality control procedure in operational remote sensing surveys would enable a more reliable retrieval of the variables of interest.

The bathymetry retrieved by applying the integrated physics based mapping approach to airborne hyperspectral data was compared to the bathymetry estimated using a vessel based acoustic bathymetry acquired during a high resolution multi-beam survey within 2 months of the image acquisition. The results show that the agreement between the two datasets varies as a function of the contribution of substratum to the remote sensing signal. As expected, there is greater agreement in shallower clear water than deeper or more turbid water. The objective and model-based quantitative identification and screening of the "optically deep waters" and the "quasi-optically deep waters" led to improved precision in the depth retrieval from remote sensing imagery. With respect to the airborne hyperspectral datasets, the precision of the bathymetry is a function of the substratum detectabilty (SD), while the accuracy is a function of the calibration of the hyperspectral imagery, the parameterization of the atmospheric correction and the parameterization of the model adopted for the retrieval.

The adopted approach represents a significant improvement over commonly used approaches for estimating depth from airborne and satellite multi-spectral images as it allows to overcome the degradation of accuracy in the depth retrievals as the waters become deeper or more turbid. Further, application of this integrated physics based mapping approach will enable mapping and detection of changes in bathymetry or substratum cover type on airborne and satellite multi- and hyper-spectral image data.

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